

**WHAT MAKETH THE IPHONE5??**

Anand Lal Shimpi dissects the system-on-chip inside the iPhone 5 - the A6 - which is an Apple designed SoC. The article clearly is meant for the uber-geek for whom words such as ARM, clock speeds, battery life, performance are music to the ears. Read on:

<http://dgit.in/SAue5x>

**POWERFUL PINNERS**

For all those Pinterest fanatics out there, this is a

definitive list of 51 powerful pinners from all across the world, who have in excess of a million followers. As expected 90 percent pinners are women.

<http://dgit.in/VbILIE>

**DIY CAMERA BAG**

Feel that camera bags are too expensive? How about making your

own camera-carrying bag?

This tutorial gives a step-by-step method to make your own camera bag. Some of you may require your Mom's help with the sewing bit.

<http://dgit.in/NB0p69>

**A PILOT-EYE VIEW**

We have all seen videos from the side-windows of air-planes. Ever wondered what it is like to see things from a pilot's perspective? This video gives a lovely view of the scene from a pilot's seat:

<http://dgit.in/Pzo2fv>

# How hackers will jailbreak the iPhone 5?

- By Sebastian Anthony

**H**ow does a hacker jailbreak an iPhone or iPad, though? Well, I'm glad you asked, because the answer is rather interesting.

**Defining the problem**

To begin with, hackers aren't interested in hacking the iPhone 5 itself — they're actually looking for a flaw in iOS 6 and the A6 SoC, both of which are brand new and relatively unknown. In the iPhone 4S's case, it withstood hacking attempts for months — much longer than any other Apple device — before it finally fell.

To create an untethered jailbreak for the iPhone 5, hackers will first have to find an exploit in the iOS 6 kernel, and then they'll have to work out a way of circumventing the hardware-level security provided by the SoC so that they can inject arbitrary, unsigned code into the boot ROM — the first code that is executed when an iDevice is powered on. This custom code will disable the iDevice's security features, allowing you to install non-App Store programs, such as Cydia. Voila, one jailbroken iPhone.

**Finding a kernel exploit**

On something like a Linux PC, where you have full access to the source code and the ports on the back of the computer, finding a kernel exploit is relatively easy — it's just a case of painstaking analysis, leaving no stone unturned. iOS's source code is closed, however (though XNU, which it is based on, is open source), and the hardware is relatively locked down.

In the case of iOS 4 and 5, both of which have been jailbroken, the kernel has a built-

in debugger — a tool that spits out a lot of information about the kernel's behavior, so that Apple's internal software team can find and squash bugs. This debugger is only accessible via serial connection, however — and obviously, the iPhone doesn't have a serial connector on the bottom. Or does it?



It turns out that the old 30-pin Apple connector actually has two pins set aside for serial communications — and to use them, all you have to do is solder together a few simple components that can be bought for around \$30.

With the home-brew cable made, an Apple hacker can open a serial connection with the iDevice, gaining access to the kernel debugger. Once you have access to the kernel debugger, it's a matter of finding an exploit — a flaw in the kernel that can be used to gain root access to the device. This step is incredibly complicated, requiring a vast amount of software expertise. For more info, hit up Stefan Esser's excellent Black Hat and CanSecWest [PDF] presentations on iOS kernel exploitation.

**Tethered or untethered?**

Once you've found a kernel exploit and gained root access, you have achieved a tethered jailbreak. If the hacker can also find a vulnerability in the device's hardware-level security (as Limer1n did with A4-based iDevices), then the exploit can be loaded into the boot ROM and executed every time the device is powered on — an untethered jailbreak.

In the case of Apple's A5 SoC, which debuted in the iPad 2 in March 2011, it took ten months to find an exploit that would allow an untethered jailbreak. In the words of a Chronic Dev Team spokesperson: "I don't know if any iOS hacker anticipated how much the A5 chip would completely change the game & up the stakes. The endless war we fight to jailbreak has become more & more difficult with each new device released, and our recent battle against A5 only proved this further."

**Jailbreaking the iPhone 5 and A6 SoC**

The iPad 2 and iPhone 4S, powered by the A5 SoC, were by far the hardest iDevices to crack — previous devices usually only lasted a few days or weeks. This was partly because Apple is continually working to thwart would-be hackers — and also because Apple hired Nicholas Allegra (aka Comex), one of the key members of the iDevice hacking community. Not only did this slow down the jailbreaking of the A5, but more importantly Comex will have spent the last year hardening the A6 SoC against as many attack vectors as possible.

To read more, go here: <http://dgit.in/VBVBAAd>